A Sample Article Using IEEEtran.cls for IEEE Journals and Transactions

IEEE Publication Technology, *Staff, IEEE,*

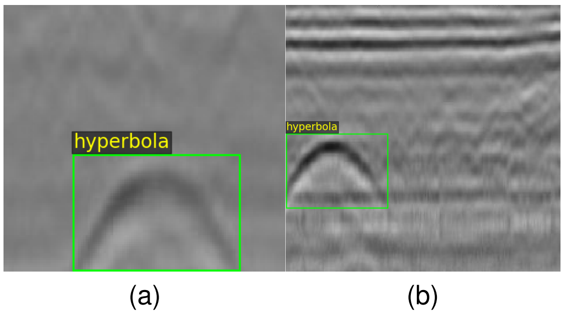

Fig. 1. Hyperbola samples.

***Abstract*—This document describes the most common article elements and how to use the IEEEtran class with LATEX to produce files that are suitable for submission to the IEEE. IEEEtran can produce conference, journal, and technical note (correspondence) papers with a suitable choice of class options.**

***Index Terms*—Article submission, IEEE, IEEEtran, journal, LATEX, paper, template, typesetting.**

I. Introduction

THIS file is intended to serve as a "sample article file"

II. Related work

A. Hyperbola recognition task

B. Labelling tools

III. Proposed method

A. Hyperbola annotation tool

We designed a visual annotation tool for hyperbolic ribbon-shaped targets, whose user interface is illustrated in the Fig. 2. First, the user selects an image folder, and the tool automatically loads all images within it. The original image is displayed on the lower-left panel, while the annotated image is shown on the right. A mouse click on the original image marks the vertex position of the hyperbola, and five adjustable parameters are provided on the right panel.

The five parameters are defined as follows: the horizontal and vertical coordinates of the hyperbola vertex, hyperbola width, height, and ribbon thickness. By clicking the *Add as New Annotation* button, the current parameter configuration is saved as a new annotation for the active image, and multiple annotations can be added to a single image.

This paper was produced by the IEEE Publication Technology Group. They are in Piscataway, NJ.

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All annotations are stored in JSON format with the following structure:

```
{
  "024.jpg": [
    {
      "label": "hyperbola",
      "x_vertex": 127.04,
      "y_vertex": 163.84,
      "width": 87.65,
      "height": 59.76,
      "thickness": 33.03
    },
    {
      "label": "hyperbola",
      "x_vertex": 186.56,
      "y_vertex": 109.76,
      "width": 97.35,
      "height": 79.35,
      "thickness": 33.03
    }
  ]
}
```

Here, *024.jpg* denotes the image filename, and the remaining entries represent the parameters and their corresponding values.

B. Hyperbola detection nerual network

IV. Experiment

V. Result and evaluation

A. Dataset

The experiments are conducted on a publicly available GPR dataset for intelligent recognition of subsurface utilities [1]. The dataset consists of labeled GPR B-scan images collected from real urban and industrial environments using ground-coupled antennas operating at 200 MHz and 400 MHz.

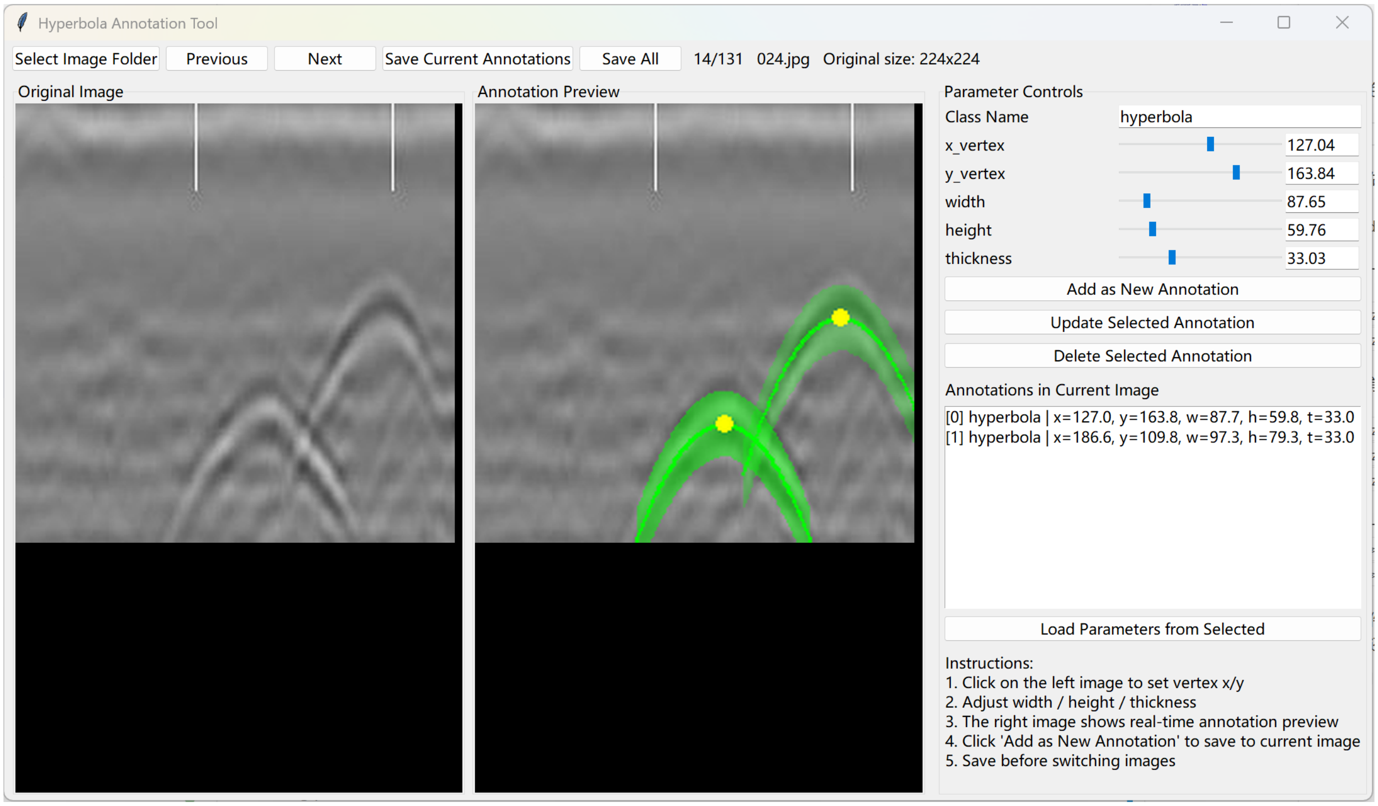


Fig. 2. tool

TABLE I
DETECTION PERFORMANCE (MEAN ± STD OVER 10 SEEDS) AT DIFFERENT TRAINING-DATA FRACTIONS. "ABS" USES THE ATTENTION CONSTRAINT WITH $\lambda_{att} = 0.7$.

Frac.	**none**			**abs** (λ=0.7)		
	P	**R**	**F1**	**P**	**R**	**F1**
0.5	0.811 ± 0.200	0.425 ± 0.094	0.529 ± 0.075	0.855 ± 0.082	0.487 ± 0.070	$\mathbf{0.617 \pm 0.063}$
0.6	0.711 ± 0.213	0.560 ± 0.083	0.600 ± 0.098	0.754 ± 0.254	0.645 ± 0.064	$\mathbf{0.663 \pm 0.131}$
0.7	0.752 ± 0.304	0.679 ± 0.097	0.678 ± 0.207	0.905 ± 0.079	0.782 ± 0.046	$\mathbf{0.836 \pm 0.032}$
0.8	0.843 ± 0.223	0.821 ± 0.071	0.811 ± 0.159	0.909 ± 0.122	0.882 ± 0.041	$\mathbf{0.890 \pm 0.069}$
0.9	0.875 ± 0.238	0.900 ± 0.068	0.867 ± 0.172	0.847 ± 0.223	0.931 ± 0.044	$\mathbf{0.870 \pm 0.165}$

config	bbox_P	bbox_R	bbox_F1
None	0.843 ± 0.223	0.821 ± 0.071	0.811 ± 0.159
se	0.8559 ± 0.1948	0.8164 ± 0.0417	0.8205 ± 0.1123
cbam	0.7478 ± 0.1649	0.8186 ± 0.0290	0.7709 ± 0.0966
nonlocal	0.8438 ± 0.2818	0.7361 ± 0.2245	0.7825 ± 0.2510
coord	0.8017 ± 0.2339	0.8355 ± 0.0904	0.7979 ± 0.1722
Ours	0.909 ± 0.122	0.882 ± 0.041	$\mathbf{0.890 \pm 0.069}$

TABLE II
COMPARISON OF DETECTION PERFORMANCE WITH DIFFERENT ATTENTION MODULES

B. Result1

C. Result2

D. Result3

E. Result4

F. Result3

G. Result3

H. Result3

VI. DISCUSSION

ACKNOWLEDGMENTS

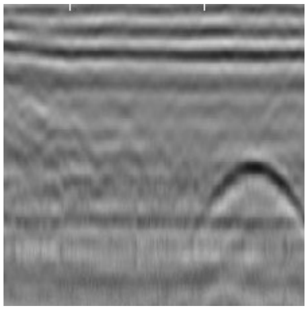
This should be a simple paragraph before the References to your work on this article.

APPENDIX
PROOF OF THE ZONKLAR EQUATIONS

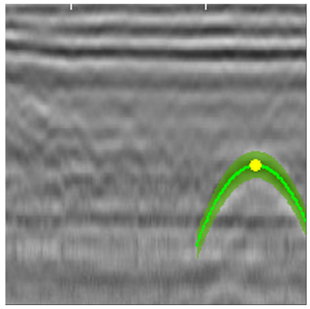
Use `\appendix` if you have a single appendix: Do not use `\section` anymore after `\appendix`, only `\section*`. If you have multiple appendixes use `\appendices` then use `\section` to start each appendix. You must declare a `\section` before using any `\subsection` or using `\label` (`\appendices` by itself starts a section numbered zero.)

REFERENCES

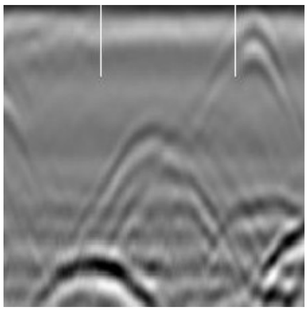
[1] A. Mojahid, D. E. Ouai, K. E. Amraoui, K. El-Hami, and H. Aitbenamer, "Intelligent recognition of subsurface utilities and voids: A ground penetrating radar dataset for deep learning applications," *Data in Brief*, vol. 59, p. 111338, 2025.



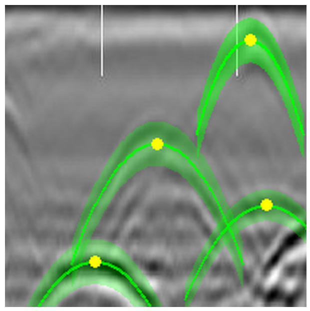
(a) Raw single-target sample.



(b) Annotated single-target sample.



(c) Raw multi-target sample.



(d) Annotated multi-target sample.

Fig. 3. Examples of hyperbola samples and their labeled visualizations.

Biography Section

If you have an EPS/PDF photo (graphicx package needed), extra braces are needed around the contents of the optional argument to biography to prevent the LaTeX parser from getting confused when it sees the complicated `\includegraphics` command within an optional argument. (You can create your own custom macro containing the `\includegraphics` command to make things simpler here.)

If you include a photo:

Michael Shell Use `\begin{IEEEbiography}` and then for the 1st argument use `\includegraphics` to declare and link the author photo. Use the author name as the 3rd argument followed by the biography text.

If you will not include a photo:

John Doe Use `\begin{IEEEbiographynophoto}` and the author name as the argument followed by the biography text.